Bearings

Year 4 · **Homework Worksheet**

Worked Example

Bearings Practice: The bearing of a buoy from a harbour is 118°. Find the back bearing of the harbour from the buoy.

Answer: Since 118° is less than 180°, add 180°: 118° + 180° = 298°. The back bearing is 298°.

Bearings Basics (15 questions)

1. What is the bearing of South, as a three-figure bearing?
2. What is the bearing of West, as a three-figure bearing?
3. Write 5° as a three-figure bearing.
4. Write 75° as a three-figure bearing. A) 7.5 B) 75 C) 075 D) 750 E) 0075
5. A boat sails on a bearing of 000°. In which compass direction is it travelling?
6. What is the bearing of North, as a three-figure bearing?
7. A car travels on a bearing of 270°. In which compass direction is it travelling? A) North B) East C) South D) West E) South-West
8. List the 8 compass directions in anticlockwise order starting from North.
9. What is the approximate bearing of South-West (SW)?
10. A cyclist is told to head on a bearing of 045°. Which compass direction does this match? A) North-East B) South-East C) South-West D) North-West E) East
11. Select ALL that apply: which of the following are valid three-figure bearings? A) 090° B) 12° C) 005° D) 400° E) 299°
12. What is the bearing of East, as a three-figure bearing?
13. A plane flies on a bearing of 000° and then turns to fly on a bearing of 180°. What is the relationship between these two directions?
14. Select ALL that apply: which of these bearings point in a generally northerly direction (less than 090° or greater than 270°)? A) 030° B) 100° C) 200° D) 320° E) 355°
15. A compass needle points on a bearing of 180°. Which compass direction is this? A) North B) East C) South D) West E) South-East

Reading & Finding Bearings (15 questions)

1. A ship's radar shows an island at a bearing of 220° from the ship. Describe, using a compass direction, roughly where the island lies relative to the ship.
2. The bearing of village B from village A is 035°. What is the back bearing (the bearing of A from B)?
3. The bearing of a tower from a car park is 198°. What is the back bearing of the car park from the tower?
4. A bearing of 080° is given. What is the back bearing? A) 240° B) 250° C) 260° D) 270° E) 280°
5. A boat sails from harbour P to harbour Q on a bearing of 155°. What is the bearing from Q back to P?

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6. Select ALL that apply: which of these pairs correctly show a bearing and its exact back bearing? A) 010° and 190° B) 100° and 190° C) 215° and 035° D) 330° and 150° E) 060° and 260°

7. A bearing of 112° is measured from North, turning clockwise. Through how many degrees would you turn, starting by facing North, to face along this bearing?

8. Two signposts show that Oakfield is on a bearing of 302° from Mill Lane. What direction, roughly (to the nearest compass point of N, NE, E, SE, S, SW, W, NW), is Oakfield from Mill Lane? A) North B) North-East C) North-West D) West E) South-West

9. A bearing of 340° is given. What is the back bearing?

10. A plane flies on a bearing of 162° from airport X to airport Y. Describe the back bearing in words (using a compass direction) and give its value as a three-figure bearing.

11. A bearing of 005° is given. What is the back bearing? A) 175° B) 180° C) 185° D) 190° E) 195°

12. Select ALL that apply: a ship travels from a buoy to a jetty on a bearing of 205°. Which statements about the return journey (jetty to buoy) are true? A) The back bearing is 025°. B) The back bearing is found by adding 180° to 205°. C) The back bearing is found by subtracting 180° from 205°. D) The return journey is on a bearing of less than 090°. E) 205° and 025° differ by exactly 180°.

13. A bearing from a ship to a buoy is measured as 325°. What is the bearing from the buoy back to the ship?

14. On a map, a farmhouse is due South-East of a barn. Give the bearing of the farmhouse from the barn.

15. A diagram shows two points, E and F. The bearing of F from E is 088°. State the bearing of E from F.

Standard / Advanced (15 questions)

1. A ferry crosses from port A to port B on a bearing of 100°. What bearing does it return on, and which compass direction is that closest to?

2. Two walkers leave the same campsite at the same time: Walker 1 on a bearing of 040°, Walker 2 on a bearing of 190°. What is the angle between their two paths, and is this angle more or less than a right angle?

3. A ship sails on a bearing of 115° for a while, then turns through 90° clockwise. What bearing is it on now, and which compass direction is that closest to?

4. A surveyor records a bearing of 140°, but later realises the true reading is 25° more than this. What is the corrected bearing, and what is its back bearing?

5. The bearing from a tower to a bridge is 072°. What is the back bearing, and is it more or less than 270°?

6. Select ALL that apply: a footpath runs from a barn to a pond on a bearing of 210°. Which of these statements are true? A) The pond-to-barn back bearing is 030°. B) The pond lies roughly north-east of the barn. C) To find the back bearing here, you subtract 180° from 210°. D) The barn lies roughly north-east of the pond. E) The bearing and its back bearing are exactly 180° apart.

7. Two ships leave the same point: one on a bearing of 040°, the other on a bearing of 310°. What is the angle between their two bearings? A) 70° B) 80° C) 90° D) 100°

8. Flying from A to B uses a bearing of 100°; flying on from B to C uses a bearing of 280°. What does this tell us about A and C? A) They are the same point. B) C is north of A. C) C is south of A. D) There is not enough information to say. E) A and C cannot both exist.

9. Looking out from a lighthouse, the keeper spots a yacht at a bearing of 075°. What bearing would someone aboard that yacht use to point back at the lighthouse?

10. Select ALL that apply: a train's bearing from Town M to Town N is 152°. Which of the following are correct ways to describe the return bearing from N to M? A) 152° − 180° = −028°, so add 360° to get 332°. B) 332° C) 180° − 152° = 028° D) The return bearing and the outward bearing differ by 180°. E) 152° + 180° = 332°.

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11. Having left harbour on a bearing of 135°, a ship reverses course to head straight back. What bearing does it now follow, and which compass direction does that match most closely? A) North B) North-East C) South-East D) South E) North-West

12. Starting from a bearing of 162°, add 180° to find the back bearing. Which compass point out of N, NE, E, SE, S, SW, W, NW is closest to that result?

13. A ship sails on a bearing of 100° for 12 km to point B, then on a bearing of 280° for 5 km to point C. How far is C from the ship's original starting point?

14. The outward bearing from airport R to airport S is 065°. A student claims the back bearing from S to R is 245°. Show that the student is correct by working out 065° + 180°.

15. The outward bearing from airport R to airport S is 095°. What is the back bearing from S to R, and by how many degrees do the outward and back bearings differ?

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